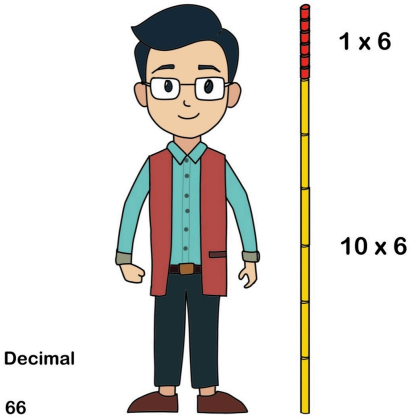
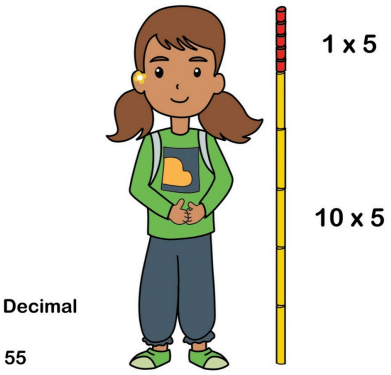
Measure your Height

Grade 8: Story of Numbers

Introduction

We will explore the concept of the positional number system through hands-on activities. We will develop an intuition for various base representations of numbers by measuring students' heights, recognising that decimal numbers that we learn in school are just one way to represent numbers.





Activity	Time	Description
Launch	5 min	The teacher poses the question to the students and invites a couple of students to try their ideas. Activity Video Link: https://tinyurl.com/heightactivityvideo
Problem-solving and algorithm	25 mins	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/heightworksheet As the teacher progresses through the activities, the students progress along with the worksheet.
Discussions and Explorations	5 min	The discussion on the various algorithms, how a slight change in approach leads to a faster solution, and how computers apply such algorithms

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CT Connection

Pattern recognition: Two see that the same patterns of representation appear in each of the base systems and the symbols used in them.

Algorithmic Thinking: We use a step-by-step approach to measure the height using the decimal, binary, and ternary pipes.

Decomposition: Breaking the conversion of the number to binary and ternary into one pipe at a time.

Abstraction: Two use the pattern seen in decimal and apply it to powers of two and powers of three.

Generalisation: Recognising a doubling pattern of 1, 2, 4, and so on, and adding different powers thus generated yields the binary representation; we conclude that any number can be represented in terms of powers of 2. Similarly, we see how to measure using ternary numbers. This method can thus be generalised for any natural number n.

Logic: To understand why the digits start from zero and end before the base number n (0 to $n - 1$)

Activity: Measuring your Height

1: Measuring your height with Decimal Pipes

We generally measure our height in the decimal number system, i.e., using base -10 numbers. We use powers of ten 10^0, 10^1, 10^2, 10^3, … (1, 10, 100, 1000, …) and digits from 0 to 9 (ten distinct symbols), to represent any number that exists.

Imagine now that you are measuring heights using PVC pipes of fixed lengths:
1 inch, 10 inches, 100 inches, and so on
Each pipe size is a power of 10. You also have 9 pipes of each length.



1. How many of the decimal pipes will it take to measure the height of a 66-inch-tall person?

a. 21 pipes: 5 pipes of 10 inches height and 16 pipes of 1 inch height
b. 7 pipes: 7 pipes of 10 inches height
c. 9 pipes: 6 pipes of 10 inches and 3 pipes of 2 inches height
d. 12 pipes: 6 pipes of 1 inch and 6 pipes of 10 inches

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Correct answer: d. 12 pipes: 6 pipes of 1 inch and 6 pipes of 10 inches

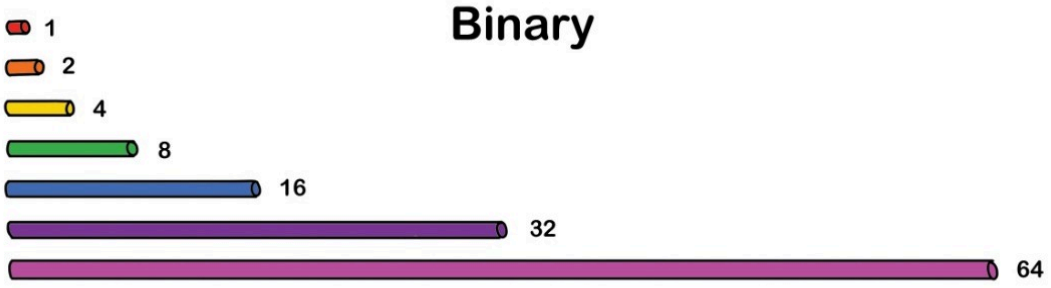
Explanation: 66 in decimal notation is represented as 6x10 + 6x1, that is, 6 of 10s and 6 of 1s, for a total of 12 pipes.

Competencies: Algorithmic thinking, logic, decomposition

2: Measuring your height with binary pipes

We keep hearing that the language of computers is binary, the language of 0 and 1. But what are these binary numbers? How do we understand these in light of what we already know?

Instead of using decimal pipes, which are powers of 10, what happens if we use pipes of length 1, 2, 4, 8, 16, 32, 64, and so on? What is special about these numbers?
They are powers of 2. We have exactly one copy of each pipe length. We will call these pipes Binary pipes.

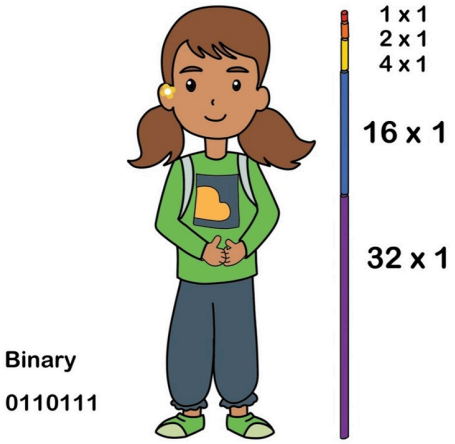


We have only one copy of each pipe for the following reason. Two 2-inch pipes make a 2-inch pipe, but we already have a 4-inch pipe. Two pipes of 4 inches make a 8 inch pipe, but we already have an 8-inch pipe; likewise, we can do this for pipes of all lengths.

Let's measure height in binary. For example, 55 = 32 + 16 + 4 + 2 + 1.
We will see that if we keep a 64-inch pipe, it's more than the height we are measuring, so we don't use a 64-inch pipe. We use the next lower power of 2, 32. We then keep 16, the next lower power. We then keep the next lower power of 2, which is 8, but this is more than the height we are measuring. So we don't use it and move on to the next power of 2. This process continues.

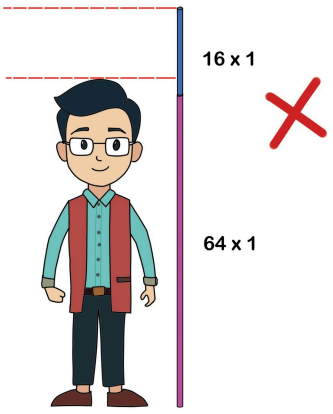
We can tabulate the information below. Using this, we can say that the binary representation of 55 is $(110111)_2$.

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55 in binary	$2^5 = 32$	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
$(110111)_2$	1	1	0	1	1	1

2. Which all binary pipes will we use to measure the height of a 66-inch-tall person?
    a. 63, 3
    b. 32, 16, 8, 4, 2, 1, 2, 1
    c. 64, 2
    d. 32, 32, 2



Correct answer: c.

Explanation: Let's understand what is happening. We first place a 64-inch-tall block. Now, when we try to keep the other heights over 64 inches, like 32, 16, 8, 4, etc., we will see that these heights will exceed the height we have to measure. So, we pick

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the next lower height until we reach the one that fits. In the number 66, we have 1 copy of 64, 0 copies of 32, 0 copies of 16, 0 copies of 8, 0 copies of 4, 1 copy of 2, and 0 copies of 1. If we write this information in the same way we write place values for decimals, we get:

$2^7 = 128$	$2^6 = 64$	$2^5 = 32$	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1	0	0	0	0	1	0

Thus, 66 in decimal in binary representation is $(1000010)_2$. We write a $(----)_2$, subscript 2 to tell us that this is a base 2 representation of a number.

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

3. Binary representation of 10 is:
    a. 1010
    b. 1111
    c. 1000
    d. 1100

Correct answer: a.

Explanation: We can draw a table similar to the one shown above and list the powers of 2 that add up to 10.

2^3	2^2	2^1	2^0
1	0	1	0

Thus, $10 = 1 \times 8 + 1 \times 2 = (1010)_2$.

Competencies: algorithmic thinking, abstraction, generalisation, pattern recognition

4. Why can't we use the digit “2” in the representation of a number in binary?
    a. It is illegal, as only 0 and 1 are given to us in the rules
    b. We can write any number using 0 and 1; there is no need for 2
    c. Computers dislike it
    d. It is too big

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Correct answer: b.
Explanation: If we used the digit “2” in the binary system in the ones place, it would denote 2 x 1 inch = 2 inches. But this is equivalent to using one 2-inch block, which is already present among the height blocks. This means redundancy.

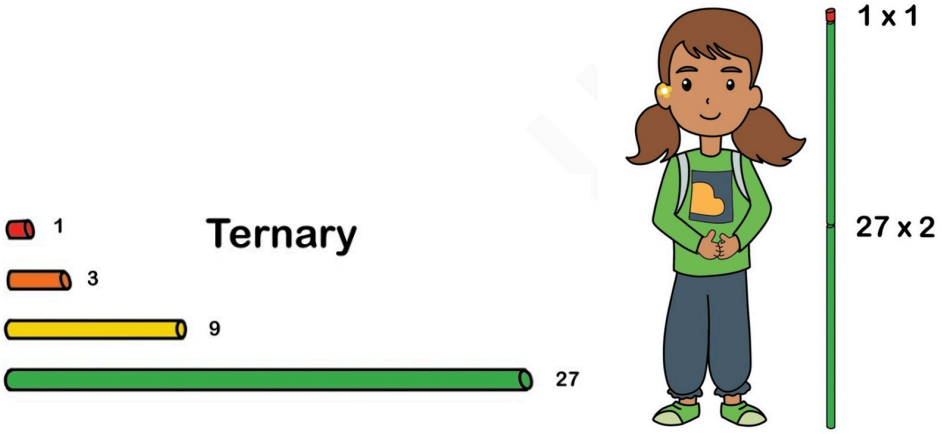
That is why digits must be less than the base. Similarly, two blocks of 4 inches make 8 inches, but we already have an 8-inch block; likewise, two 8 inches is 16 inches, and there is already an 16 inches block

Competencies: abstraction, generalisation, pattern recognition, logic

3: Measuring your height with ternary pipes

Instead of base 10 (decimal) and base 2 (binary), what if we set the base to something else, for example, 3?

Then we will have pipes of lengths 1, 3, 9, 27, 81…, all powers of 3, and 2 pipes of each length.



Let’s see an example of measuring 55 inches in height in ternary.

We will see that if we keep the 81-inch pipe, it's taller than the height we are measuring, so we don’t use it. We use the next lower power of 3, 27. We keep two of these, which is the maximum number of pipes of each type allowed.
We then keep 2 pipes of 9, the next lower power, but this overshoots the height, so we remove one pipe. Even with one 9-inch pipe, we are still over the height we are measuring.
So remove this pipe and keep two 3-inch pipes. But this overshoots the height, so we remove one pipe. Even with one 3-inch pipe, we are still over the height we are measuring.
So, we remove that and keep two pipes of the next lower power of 3, which is 1, but this is more than the height we are measuring. So, we remove one pipe and see that we get the correct height measurement.
55 inches is $54 + 1 = 2 \times 27 + 1 \times 1$.

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55	3^3	3^2	3^1	3^0
$(2001)_3$	2	0	0	1

5. When we measure 66 inches with ternary pipes, how will its representation be?
    a. 2011
    b. 2101
    c. 2110
    d. 2210

Correct answer: c.

Explanation: To measure 66 inches, we use two 27-inch pipes, one 9-inch pipe, one 3-inch pipe, and zero 1-inch pipes.
$66 = 2\times27 + 1\times9 + 1\times3 + 0\times1 = (2110)_3$

66	3^3	3^2	3^1	3^0
$(2110)_3$	2	1	1	0

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

6. What is the largest digit allowed in Base 3?
    a. 3
    b. 2
    c. 1
    d. 9

Correct answer: b
.
Explanation: As we saw in other representations, the digits allowed in a base number system are numbers less than the base. In case 3, we have three numbers: 0, 1, and 2.

Competencies: abstraction, generalisation, pattern recognition, logic

7. How will we write 100 in base 3 (ternary) notation?
    a. $(3201)_3$
    b. $(2333)_3$

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c. $(11000)_3$

d. $(10201)_3$

Correct answer: d.

Explanation: $100 = 1 \times 81 + 0 \times 27 + 2 \times 9 + 0 \times 3 + 1 \times 1 = (10201)_3$

100	3^4	3^3	3^2	3^1	3^0
$(10201)_3$	1	0	2	0	1

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

Conclusion

We see that a number can be represented in many forms and symbols. Decimal notation is one of the many possible base number systems (Positional value system).

The fundamental idea is that we arrange the powers of the base number n in increasing order from right to left.

The face value at each place tells us how many times the corresponding power of the base is used to represent the number.

We also see that the base n system requires n distinct digits. These digits are 0 till $n - 1$.

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